Siavash Shami

M.Sc. in Geodesy | InSAR & Remote Sensing Researcher
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Research Profile

I am a geodesist and InSAR researcher specializing in SAR-based surface deformation monitoring and Earth observation. My research focuses on InSAR methodology, interferometric time-series analysis, network design, and error characterization, with an emphasis on reliable deformation measurements from satellite radar observations. I integrate InSAR with GNSS, remote sensing, geological, and hydrological data to investigate natural and anthropogenic deformation. I also apply machine learning and deep learning approaches to deformation modeling and environmental remote sensing.

Research Interests

- InSAR and SAR Remote Sensing
- Surface and Crustal Deformation Monitoring
- InSAR Time-Series Analysis, Network Design, and Error Characterization
- Geohazards and Natural & Anthropogenic Deformation
- Environmental and Climate Applications of Remote Sensing
- Machine Learning and Deep Learning for Geodesy and Earth Observation

Experiences

Education

- **M.Sc. in Geodesy**
  K. N. Toosi University of Technology, Iran | 2018
  Thesis: *A Feasibility Study on the Application of InSAR for Surveying in Open Pit Mines* [In Persian]
- **B.Sc. in Surveying Engineering**
  Eyn Ol Ghozat Higher Education Institute, Iran | 2016

Professional Experiences

- Research Collaboration - InSAR Applications | 2023-2024
  [Collaborated with Dr. Meisam Amani on InSAR-based investigations of landslides in New Zealand, dam deformation and shoreline changes in Canada, and open-pit mine deformation in South Africa]
- InSAR & GIS Consultant/Research Collaborator - Bonyan Zamin Co. | 2022
  [Prepared annual InSAR-based surface displacement maps of Isfahan, Iran, for Isfahan Municipality using Sentinel-1 data, PS-InSAR, GMTSAR, StaMPS, and QGIS]
- Military Service | 2019-2021
  [Served as a full-time conscript; continued independent research and technical writing during this period and authored a practical handbook on InSAR processing]

Peer Reviewer

- Gondwana Research - Elsevier (1)
- Earth Science Informatics - Springer Nature (1)

- Geography, environment, sustainability (1)
- IEEE Access - IEEE (1)
- Advances in Space Research - Elsevier (1)
- Remote Sensing Applications: Society and Environment - Elsevier (3)
- Revue Internationale de Géomatique - Tech Science Press (1)
- Natural Hazards - Springer Nature (1)

Awards & Honors

- Ranked 1st among B.Sc. students in Surveying Engineering [GPA: 3.93/4.00] | 2016

Skills

InSAR & SAR Remote Sensing

- SAR Data: Sentinel-1, ENVISAT, ALOS-1/2, ERS-1/2
- InSAR Methods: DInSAR, PS-InSAR, SBAS, NSBAS
- InSAR Processing: Multilooking, noise reduction, interferogram generation, network design and optimization
- Error Characterization & Correction: Atmospheric correction (ERA5, GACOS), filtering, phase-bias analysis
- Software & Processing Tools: StaMPS/MTI, GMTSAR, MintPy, ISCE, LiCSBAS, SNAP, DORIS, ROI_PAC

Geodesy & Geodetic Techniques

- Satellite Geodesy and Physical Geodesy
- GNSS: Dual-Frequency GNSS
- Levelling and Topographic Surveying
- Total Station and Nivo
- Surveying & Geospatial Software: Civil 3D, Global Mapper, RTKLIB

Remote Sensing & Environmental Applications

- Applications: Soil Moisture, Groundwater Storage, Precipitation, Land Cover, Air Pollution, Climate Change, Soil Erosion
- Data & Products: Landsat, Sentinel-1/2, MODIS, GRACE, Well Data, CHIRPS, TRMM, Sentinel-5P/TROPOMI

AI & Data Analysis

- Machine Learning: Random Forest, XGBoost, SVM, Linear Regression, Lasso Regression
- Deep Learning: RNN, LSTM

Programming & GIS

- Programming & Data Analysis: Python, NumPy, Pandas, Matplotlib
- GIS & Remote Sensing Platforms: QGIS, Google Earth Engine
- Operating Systems: Linux (Ubuntu)

Languages

- Azerbaijani: Native
- Persian: Native
- English: Professional Working Proficiency

Publications

18. Naboureh, A., **Shami, S**., Zamani, A., & Bateni, S. M. (2027). Variation in RUSLE-estimated soil erosion responses to land use/land cover change across major climate zone. *Catena*
Co-first author (equal contribution)
Research Article - Peer Reviewed
DOI: 10.1016/j.catena.2026.110718

17. Karami, E., **Shami, S**., Maghsoudi, Y., Ranjgar, B., & Azadnejad, S. (2025). Investigating the InSAR Phase Bias in the SBAS Algorithm and Its effect on different Landcovers. *IEEE Access*
Research Article - Peer Reviewed
DOI: 10.1109/ACCESS.2025.3568144

16. Naboureh, A., Li, A., Bian, J., Lei, G., Nan, X., Zhang, Z., **Shami, S.** & Lin, X. (2024). Green space coverage versus air pollution: a cloud-based remote sensing data analysis in Sichuan, Western China. *International Journal of Digital Earth*
Research Article - Peer Reviewed
DOI: 10.1080/17538947.2024.2383454

15. **Shami, S.**, Shahriari, M. A., Nilfouroushan, F., Forghani, N., Salimi, M., & Reshadi, M. A. M. (2024). Surface displacement measurement and modeling of the Shah-Gheyb salt dome in southern Iran using InSAR and machine learning techniques. *International Journal of Applied Earth Observation and Geoinformation*
Research Article - Peer Reviewed
DOI: 10.1016/j.jag.2024.104016

14. Abdalla, A., **Shami, S.**, Shahriari, M. A., & Azar, M. K. (2024). Estimation of land displacement in East Baton Rouge Parish, Louisiana, using InSAR: Comparisons with GNSS and machine learning models. *The Egyptian Journal of Remote Sensing and Space Sciences*
Co-first author (equal contribution)
Research Article - Peer Reviewed
DOI: 10.1016/j.ejrs.2024.02.008

13. Khoshlahjeh Azar, M., **Shami, S.**, Nilfouroushan, F., Salimi, M., Ghayoor Bolorfroshan, M., & Reshadi, M. A. M. (2022). Integrated analysis of Hashtgerd plain deformation, using Sentinel-1 SAR, geological and hydrological data. *Scientific Reports*
Co-first author (equal contribution)
Research Article - Peer Reviewed
DOI: 10.1038/s41598-022-25659-4

12. **Shami, S.**, Azar, M. K., Nilfouroushan, F., Salimi, M., & Reshadi, M. A. M. (2022). Assessments of ground subsidence along the railway in the Kashan plain, Iran, using Sentinel-1 data and NSBAS algorithm. *International Journal of Applied Earth Observation and Geoinformation*
Research Article - Peer Reviewed
DOI: 10.1016/j.jag.2022.102898

11. **Shami, S.**, Ranjgar, B., Bian, J., Khoshlahjeh Azar, M., Moghimi, A., Amani, M., & Naboureh, A. (2022). Trends of CO and NO2 Pollutants in Iran during COVID-19 pandemic using Timeseries Sentinel-5 images in Google Earth Engine. *Pollutants*
Research Article - Peer Reviewed
DOI: 10.3390/pollutants2020012

10. **Shami, S.** (2021). GMTSAR installation and processing Guide - Practical InSAR handbook series. *ebook*
Book - Handbook
DOI: 10.5281/zenodo.22609191

9. **Shami, S.**, Khoshlahjeh, M., Ghorbani, Z., Moghimi, A., Mohammadzadeh, A., & Sabet Ghadam, S. S. (2021). Evaluation of air pollution contributes for the COVID-19 pandemic in Iran using Sentinel 5 Satellite Data. *Journal of Geomatics Science and Technology* [In Persian]
Research Article - Peer Reviewed

8. Hossainali, MM., & **Shami, S.** (2020). Processing Radar Images using StaMPS. *K.N. TOOSI UNIVERSITY Press* [In Persian]
Co-first author (equal contribution)
ISBN: 978-622-6655-46-0
Book - Technical Book

7. **Shami, S.**, & Ghorbani, Z. (2019). Processing Radar Images using GMTSAR & SNAP. *Arshadan Press* [In Persian]
ISBN: 978-622-251-021-3
Book - Technical Book

6. **Shami, S.**, & Ghorbani, Z. (2019). Investigating water storage changes in Iran using grace and chirps data in the google earth engine system. *The International Archives of the Photogrammetry, Remote Sensing and Spatial Information Sciences*
Conference Paper - Poster Presentation
DOI: 10.5194/isprs-archives-XLII-4-W18-981-2019

5. **Shami, S.**, & Ghorbani, Z. (2019). Influence of wavelength radar images in estimation of open-pit mine displacements using radar interferometry technology (case study: Gol Gohar Sirjan mine). *3rd TRIGGER International Conference*
Conference Paper - Poster Presentation

4. **Shami, S.**, Hossainali, MM., & Babaee, S. (2019). Analysis of Large-scale Displacement Using Radar Interferometry Technology in Open-pit Mines (Case Study: Gol Gohar Sirjan Mine). *Geospatial Engineering Journal* [In Persian]
Review Article - Peer Reviewed

3. **Shami, S.**, Ghorbani, Z., & Abedi, L. (2018). Calculation of the extracted mines volume using DEM obtained from the Envisat radar satellite images (Case study: Gol Gohar Sirjan Mine). *The 4th National Geology and Mining Explorations Symposium* [In Persian]
Conference Paper - Oral Presentation

2. Shami, S., Ghorbani, Z., & Abedi, L. (2018). Persistent Scatterers Time series analysis of radar interferometry to determine the rate subsidence of areas around the open pit mines (Case study: Gol Gohar Sirjan Mine). *The 4th National Geology and Mining Explorations Symposium* [In Persian]
Conference Paper - Poster Presentation

1. **Shami, S.**, & Hossainali, MM. (2018). Analysis of the effect of temporal and spatial baseline of radar images on the amount of coherence for detecting large-scale displacement gradients using interferometric synthetic aperture radar (Case study: Gol Gohar Sirjan Mine). *Geomatics and GIT 97* [In Persian]
Conference Paper - Poster Presentation